

An Explainable AI Approach to Algorithmic Fairness of Machine Learning Models for Alzheimer's Disease Prediction

Introduction

Alzheimer's Disease (AD) is a debilitating neurological condition predominantly affecting the elderly, characterized by a decline in cognitive abilities and daily functioning. As reported by the latest World Alzheimer Report, AD affects approximately 55 million people globally, with projections indicating a rise to 139 million by 2050 [1, 2]. Alarmingly, about 75% of cases remain undiagnosed due to various factors [3]. Patients with AD face significant challenges, including severe memory loss, behavioral changes, and impairments in vision and mobility, which profoundly disrupt their everyday lives. Mild Cognitive Impairment (MCI), a precursor to AD, progresses to dementia disease in about 10% of cases, underscoring the need for early detection and intervention [5]. The recent FDA approval of drugs such as Lecenamab and Abucanemab highlight the growing emphasis on early detection in Alzheimer's research. As the seventh leading cause of death worldwide, AD's growing prevalence calls for urgent attention to improve diagnosis, treatment, and support for affected individuals and their families.

Through the integration of machine learning techniques, artificial intelligence has strongly impacted the advancement of a broad spectrum of fields within medicine [6, 7, 8]. Among other ranging benefits from treatment arrangement to patient care, integration of these technologies has

led to diagnostic accuracy across multiple diseases [10, 11]. Considering the large availability of Alzheimer's data, researchers have employed numerous machine learning techniques for the prediction of Alzheimer's often including but not limited to k-nearest neighbors (KNN), random forest and artificial neural networks (ANN). Although machine learning (ML) algorithms have the potential to enhance clinical decision-making and improve diagnostic accuracy [12] there is an increasing concern that these algorithms might unintentionally introduce bias into decision-making processes, which could result in unintended discrimination against underrepresented and disadvantaged populations [13].

LIME (Local Interpretable Model-agnostic Explanations) is an explainable artificial intelligence (XAI) technique that explains the relative importance of features towards model decisions, variations in these values indicate changes in the models feature space and decision making process. Although it is not a traditional application, in the context of model interpretability and fairness, differences in LIME values across racial groups can be indicative of potential biases in machine learning models. Variations in the influence of specific features for different races may suggest that the model utilizes features differently depending on racial context, which could highlight underlying biases or fairness concerns. Such discrepancies can signal that the model's decision-making process might not be equally equitable across diverse groups, raising important questions about fairness and ethical implications.

In this study, I explored the algorithmic fairness of prediction models for AD and related diseases, leveraging LIME explanations. I utilized publicly available data from the National Alzheimer's Coordinating Center (NACC) to build and evaluate classifiers for different

classification tasks, including normal cognition (NC) vs. AD and a multi-class scenario involving NC, MCI, and dementia. KNN, and ANN models were developed for these tasks to assess their fairness and identify potential biases.

We had an objective of auditing the fairness of ML models for AD prediction and testing (maybe change to observing, examining) for significant differences in LIME values. The overall goals of this study were to introduce and define fairness metrics relevant to models for predicting AD progression and to illustrate how ML algorithms may be analyzed to reveal potential disparities across demographics. Analysis of the models revealed that the KNN exhibited unfairness to the false positive rate of Black or African American participants for both binary classification and multiclass tasks. Additionally, the ANN exhibited unfairness for participants within the Native American and Alaskan Native group.

Literature Review

Machine learning, an essential branch of artificial intelligence (AI), concentrates on developing systems capable of identifying patterns within large datasets, facilitating tasks such as the development of classification and prediction models[15, 16]. This field leverages methodologies from data mining, statistical analysis, and various optimization strategies to create robust models [17]. A crucial subset within machine learning is representational learning, which focuses on discovering the most effective ways to represent data to extract meaningful insights automatically [4]. Over time, numerous machine learning algorithms have been developed and refined to enhance performance for various applications, including medical diagnostics and treatment [5].

The incorporation of AI and machine learning into medical practices represents a transformative shift in healthcare delivery. These technologies utilize the insights gained from analyzing extensive datasets to develop more precise and personalized diagnostic and treatment strategies [6]. As a result, healthcare professionals can make more accurate and timely decisions regarding patient care [7]. For instance, in oncology, machine learning models are used to predict treatment responses and tailor therapies to individual patient profiles, thus optimizing therapeutic outcomes while reducing potential side effects [8]. This approach exemplifies how AI-driven insights can significantly enhance the efficacy of medical interventions.

Alzheimer's disease (AD) is a progressive neurodegenerative disease predominantly affecting the elderly, characterized by cognitive decline, memory loss, and the buildup of damaging proteins in the brain's nerve cells. AD symptoms typically begin with mild memory loss and difficulty in completing familiar tasks [14]. As the disease progresses, symptoms such as difficulty in communicating, disorientation, mood, behaviour changes, and inability to care for oneself, may become more severe [20]. The pathology of AD includes the accumulation of amyloid-beta plaques and neurofibrillary tangles composed of hyperphosphorylated tau protein, which lead to neuronal death and brain atrophy. These pathological changes typically begin in the hippocampus and spread to other brain regions, resulting in a significant decline in cognitive functioning. Mild Cognitive Impairment (MCI) is a transitional stage from Cognitively Normal (CN) to dementia, with a 10% chance of progressing to AD, the most common form of dementia [18, 21]. The exact cause of Alzheimer's disease remains unknown, but it is believed to result from a combination of genetic, environmental, and lifestyle factors. Several genes have been identified that increase the risk of developing Alzheimer's, including the APOE-e4 gene,

which is the strongest genetic risk factor [24¹]. Additionally, lifestyle factors such as lack of physical inactivity, poor diet, and and lack of cognitive engagement may also contribute to the disease's development.

MCI is a common condition characterized by memory or cognitive issues that exceed what is expected for one's age, yet are not as severe as those observed in Alzheimer's disease or other forms of dementia. In fact, in many cases conditions will improve. However, MCI lies between the expected cognitive decline of normal aging and the more serious decline of dementia and 1 to 2 out of every 10 patients above the age of 65 go on to develop dementia.

Dementia, which develops from MCI, is an umbrella term for a range of cognitive disorders characterized by a decline in memory, language, problem-solving, and other thinking skills that affect a person's ability to perform everyday activities². Unlike the prior condition, Dementia is a disease that always affects the brain³. Alzheimer's disease is the most prevalent form of dementia, however there are other disorders such as lewy-body disorder and frontotemporal dementia.

One of the most important applications of machine learning research is in healthcare, with the aim of developing automated disease prediction systems. Numerous machine learning models trained on varying data modalities such as the images, clinical-features, and speech data or the early prediction of dementia have been proposed. Ranging from traditional algorithms such as Logistic Regression and k-Nearest Neighbors to advanced techniques like Artificial Neural Networks, Random Forests, and Deep Neural Networks.⁴ For instance, James et al. had evaluated

¹ <https://pubmed.ncbi.nlm.nih.gov/22229116/>

² <https://www.mayoclinic.org/diseases-conditions/dementia/symptoms-causes/syc-20352013>

³ <https://www.alzheimers.org.uk/about-dementia/types-dementia/mild-cognitive-impairment-mci#:~:text=A%20person%20with%20MCI%20has,always%20gets%20worse%20over%20time.>

⁴ <https://link.springer.com/article/10.1007/s10916-023-01906-7>

the performance of ML algorithms for predicting the progression of dementia in memory clinic patients. According to their findings, ML algorithms outperformed humans in predicting incident all-cause dementia within two years. Using all 258 variables, the gradient-boosted trees approach had an overall accuracy of 92% .Bansal et al. presented a comparative analysis of the, J48 decision tree, Naive Bayes, Random, and multilayer perceptrons. From the results of experiments, they had concluded that j48 outperformed the rest of the ML models for the detection of dementia, showing it's efficacy for Alzheimer's prediction. [am still deciding on others] - work on citations

Despite the monumental progress of machine learning in healthcare, a large number of Computer Aided Diagnosis (CAD) methods powered by AI have limited acceptance and adoption in the medical domain due to the typical black box nature of these AI models. Therefore, to facilitate the adoption of these AI models among the medical practitioners, the models' predictions must be explainable and interpretable. The emerging field of explainable AI (XAI) aims to justify the trustworthiness of these models' predictions by implementing algorithms during or after the model development which allow for users to understand the reasoning behind decisions. [give summary of some articles including LIME]

Although ML algorithms can inform clinical decision-making and are potentially associated with improved population health, there is growing concern that ML may inadvertently introduce bias into decision-making processes, which may be associated with unintended discrimination against underrepresented and disadvantaged populations. Algorithmic bias arises when a machine learning model performs differently on different groups not due to bias present in the input data but added purely by the algorithm.⁴⁸ It is generated by choices in the algorithmic design, including choice of predictor variables, optimization function,

regularization, and loss function. Choices for each of these aspects of the algorithm may potentially bias the outcome of the algorithms.¹⁷ Additionally, self-reported demographic and health information has the potential to facilitate observational studies of AD but may introduce bias in results when groups in the cohort have different reporting approaches.⁴⁹⁻⁵¹ There has been a promise shown that XAI methods such as LIME may be applied to algorithmic bias or fairness

Although it is not a traditional application, LIME (Local Interpretable Model-agnostic Explanations) values across racial groups can be indicative of potential biases in machine learning models. Variations in the influence of specific features for different races may suggest that the model utilizes features differently depending on racial context, which could highlight underlying biases or fairness concerns. Such discrepancies can signal that the model's decision-making process might not be equally equitable across diverse groups, raising important questions about fairness and ethical implications.

Methods

Population:

The Uniform Data Set (UDS) longitudinal and neurocognitive data, was downloaded upon request from the National Alzheimer's Coordinating Center (NACC). Originally comprising 192,088 records and 778 features (as of June 2024). All enrolled participants in the Alzheimer's Disease Research Centers (ADRCs) program had annual visits to the NACC since 2005, during which clinicians completed 18 data collection forms. The forms detailed information on genetic, environmental, and lifestyle factors. Most centers also enroll volunteers with normal cognition. Most participants in the study were highly educated. NACC participants are not a statistically

based sample of the U.S. population. They are best regarded as a referral-based or volunteer case series. Therefore, NACC data do not lend themselves to estimates of the prevalence or incidence of dementia subtypes in the general U.S. population. [23]

Study Design:

In order to evaluate fairness of different demographics, variables for sex, ethnicity, and race were investigated because previous studies in the fairness literature have highlighted algorithmic bias according to these variables. The variable sex included male and Female. Ethnicity was classified as Hispanic or Latino and not Hispanic or Latino. Race included Asian, American Indian or Alaskan Native, Black or African American, Hawaiian or Other Pacific Islander, and White. The two fairness metrics equalized odds and equal opportunity were selected. Equal opportunity in the context of machine learning algorithms is achieved when the true positive rates (TPRs) or sensitivity are consistent across all demographics, ensuring that individuals who progress to Alzheimer's disease (AD) are equally likely to be correctly identified by the algorithm, regardless of their group. Equalized odds goes a step further by requiring not only equal TPRs but also equal false positive rates (FPRs) or specificity across all groups. This ensures that the algorithm is equally accurate in identifying non-progression to AD across all peoples, thereby balancing both true and false positive rates uniformly across different demographics[24].

Machine Learning Models:

The machine learning was performed using Python programming [] and the popular machine learning library: Tensorflow [link]. Four machine learning models were developed for this project. A K-Nearest Neighbors (KNN) and an Artificial Neural Network (ANN) for the tasks for binary classification between CN and AD, as well as multiclass classification between CN, MCI, Dementia and AD. K-Nearest Neighbor leverages the proximity of data points for classification, offering a robust approach to pattern recognition. Meanwhile, the large highly dimensional dataset and difficult task of Alzheimer's classification called for the Artificial Neural Networks robust ability to capture complex patterns and relationships within the data. These models have seen past success in this domain. [25, 26].

Originally cross-validation was used for model selection and evaluation. Data was randomly partitioned into 10 equal subsets. Each 10-fold cross-validation iteration used 80% of participants for training, 10% of participants for model validation, and 10% of participants for testing. However, this yielded overall lower accuracy and F1 score than splitting the dataset into 60% training, 20% cross validation and 20% testing. So the latter proceeded with. The training set was used for model fitting, the validation set for hyperparameter selection, and the test set for model performance evaluation. The data was categorized into continuous variables, binary variables, and multinomial variables. All continuous variables were z normalized using the training set to estimate the mean and SD, which were then used to z normalize validation and test sets. Binary variables were hot-one encoded, or converted into columns of 0's and 1's. For the ANN, categorical embedding was applied to multinomial variables; a technique that transforms categorical variables into continuous vector representations. These vectors are then concatenated

with other input features and passed through the layers of the deep learning model. For all models, Bayesian Optimization was applied to optimize hyperparameters. Bayesian optimization is a technique that creates a probabilistic model of the objective function and uses it to select the most promising values in the parameter space, and then iterates this process in order to find optimal parameters.

LIME:

LIME (Local Interpretable Model-agnostic Explanations) is an explainable AI technique that operates by approximating the model locally with an interpretable model, like in this case a logistic regression model, to explain individual predictions. When a prediction is made, LIME perturbs the input data slightly and observes the changes in the output. By fitting a simple, interpretable model to these perturbed samples, LIME can highlight which features most significantly influenced the prediction for that specific instance. This interpretability generates local measures for the importance of individual features. The technique was scaled to global explainability by aggregating the local LIME values. Once global explanations were developed, differences between the variables across different sections of the demographics. Then the largest differences among the LIME values in the demographics were presented. [27]. Due to time constraints the LIME values were only computed on the binary ANN classifier.

Results

Model Performance Overview:

In evaluating the model performance, both ANN models demonstrated a notable advantage, likely on the account of the large highly dimensional dataset. The multiclass ANN classifier for NC vs. MCI vs. DEM achieved an overall accuracy of 85% and an F1 score of 0.83.5, while the binary classifier for NC vs. Alzheimer's reached an accuracy of 88% and an F1 score of 0.82. In comparison, the KNN model showed a lower performance, with the multiclass classifier achieving an accuracy of 73% and an F1 score of 0.71, and the binary classifier obtaining an accuracy of 73% and an F1 score of 0.70. The superior performance of the ANN highlights its effectiveness in handling complex patterns within the dataset⁵

Sex:

For the binary classification task NC vs Alzheimer's ANN, the TPR results were both similar, around 0.75, indicating that the model satisfies equalized odds. The FPR are values also similar, both of them being close to 0.55. This indicates that the model satisfies equalized odds. This theme continued across models. The TPR and FPR values are also very similar in the KNN, being around 0.92 and 0.57. For the multiclass ANN, the TPR values were around 0.25 and the FPR values were very low, around 0. In the KNN for multiclass classification there were no irregularities or instances of the unfairness observed.

Ethnicity:

For the binary classification task NC vs Alzheimer's ANN, the TPR results were both similar, around 0.76, indicating that the model satisfies equalized odds. The FPR are values also similar, both of them being close to 0.55. This indicates that the model satisfies equalized odds. This theme continued across models. The TPR and FPR values are also very similar in the KNN, being around 0.91 and 0.56. For the multiclass ANN, the TPR values were around 0.25

and the FPR values were very low, around 0. In the KNN for multiclass classification there were no irregularities or instances of the unfairness observed.

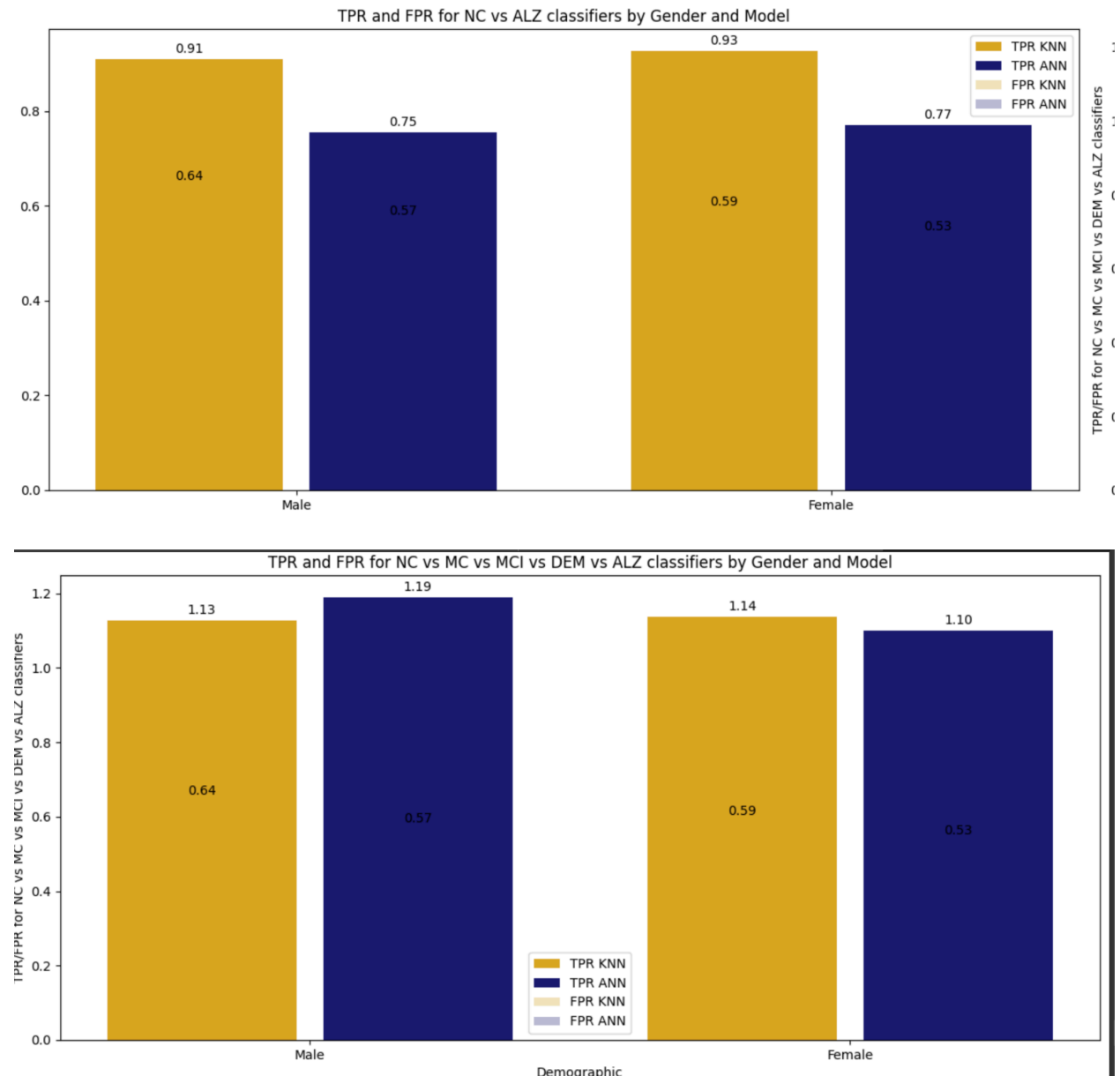
Race:

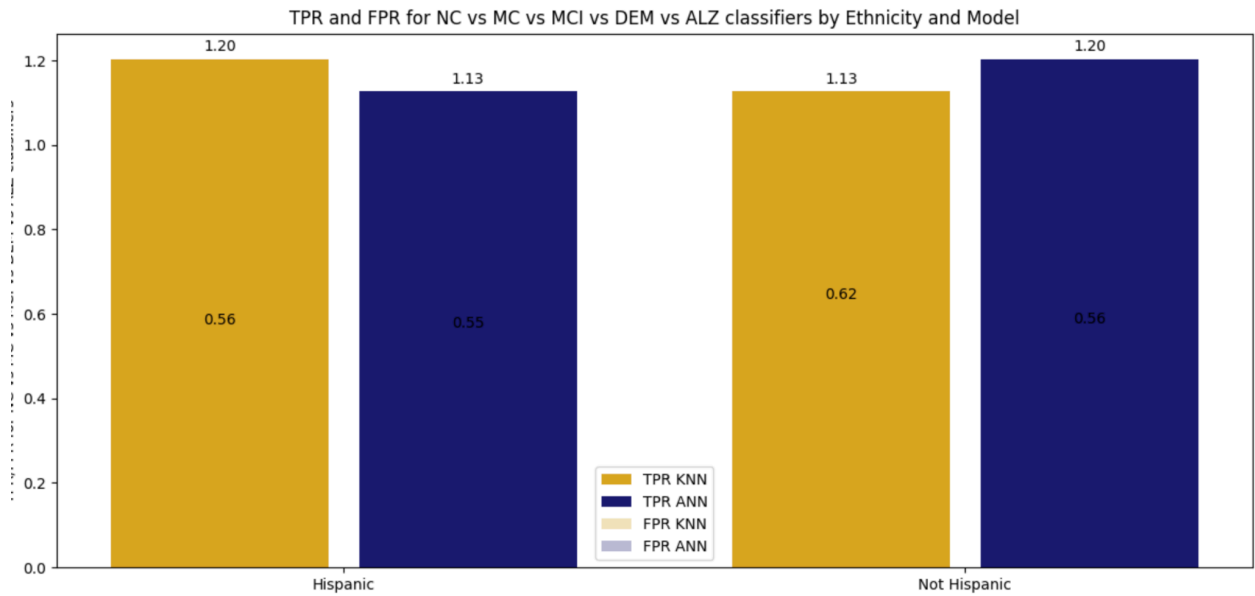
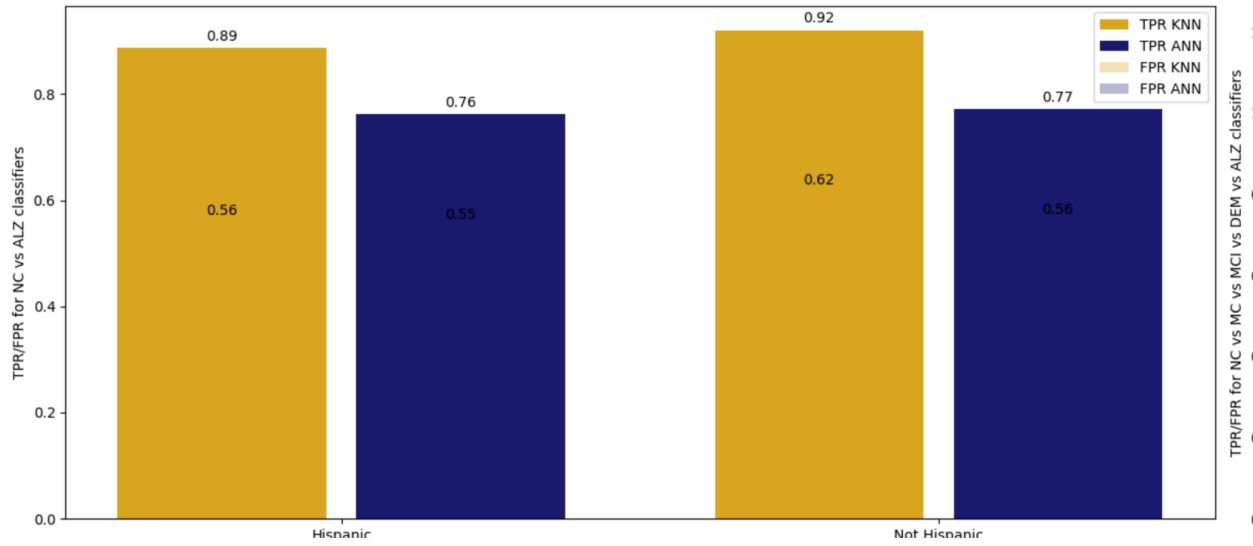
For the binary classification task NC vs Alzheimer's ANN, the TPR results were mostly similar. All of the values were around 0.75 with the exception of the Native American and Alaskan Natives, who had a higher TPR of 0.85. Additionally, Black or African American participants had a slightly higher FPR of around 0.5 rather than 0.44. Additionally Black or African American participants had a lower TPR using the binary KNN of around 0.86 as opposed 0.95 in the other categories. For the multiclass classification task, the ANN had a slightly higher TPR for Native American and Alaskan Natives, who had a higher TPR of 0.85. In the KNN for multiclass classification there were no irregularities or instances of the unfairness observed.

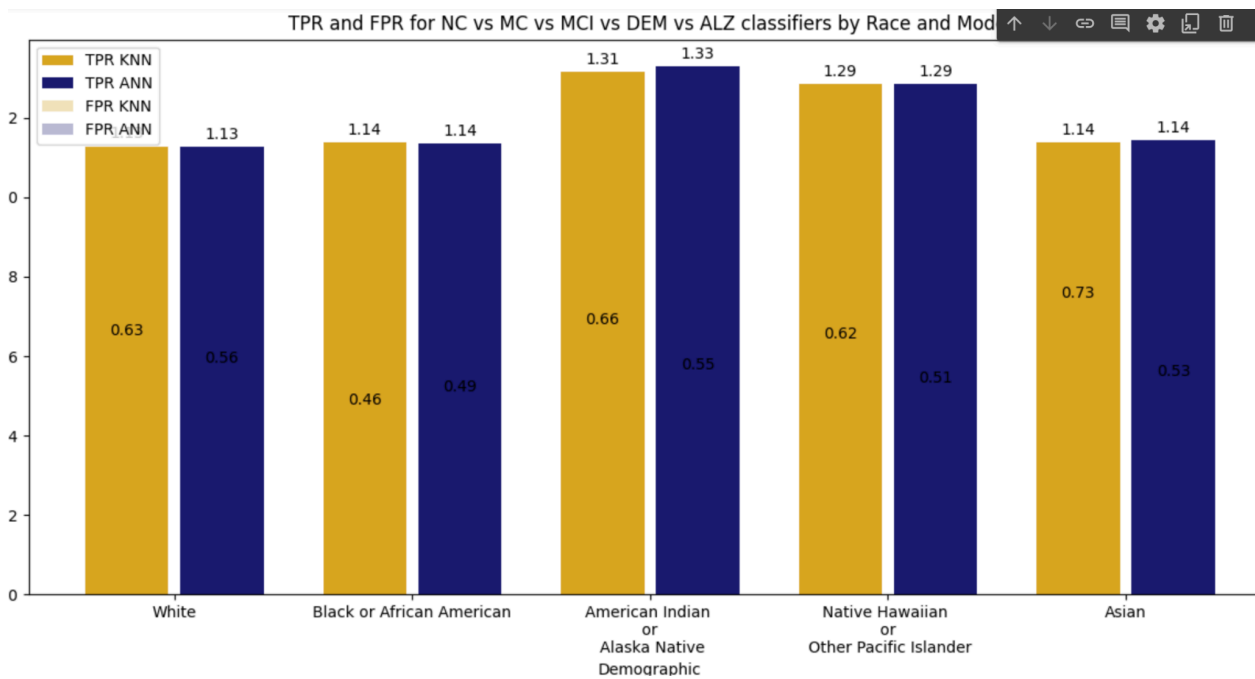
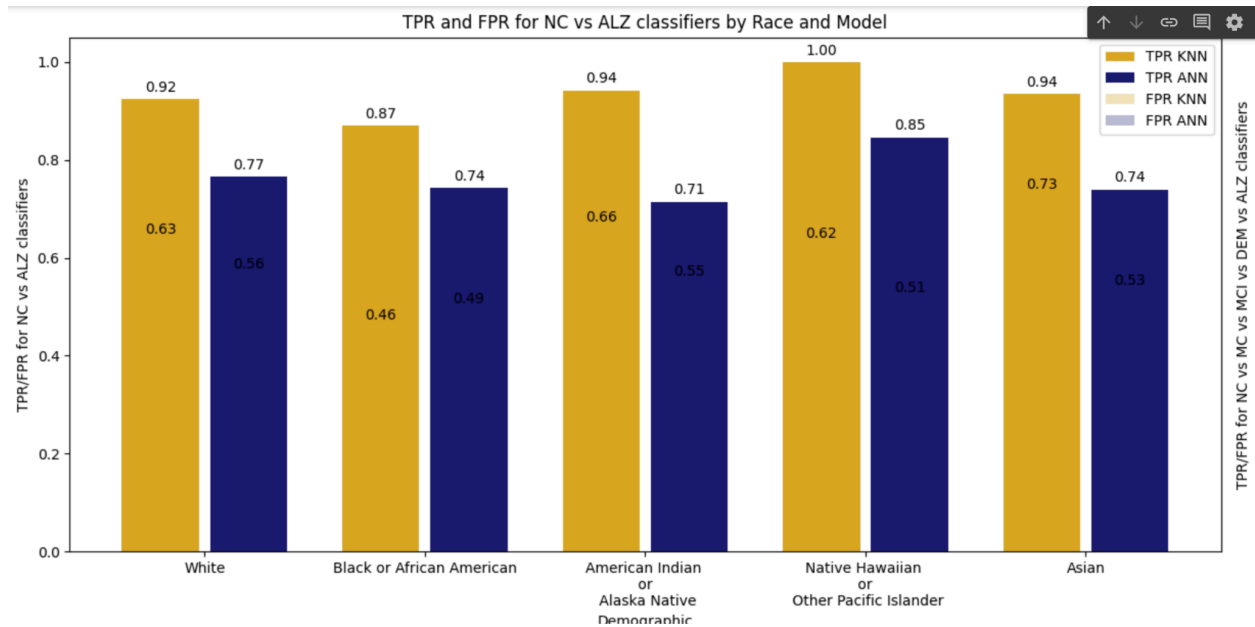
LIME Results:

The distribution of the LIME values for different groups within the demographics is linked [here](#). The largest or most significant differences within the distributions for each model have been selected. In general, the column for memory varied the most among different demographics. The difference between the importance of the memory column in male vs female participants in the binary classifier was -3.829 in males and -0.967 in females. Additionally, there was shown to be a large difference in correlation for the orientation column for men and women, for men it had a strong negative impact -1.265 and for women a smaller impact of -0.336. For the difference in the distribution of LIME values among those with and without hispanic ethnicity, there was a difference in the impact of memory, it being -2.998 for those without and -2.880 for those with. The column for personal care was also of a large difference for hispanic vs non-hispanic participants. It was more important for non-hispanic participants with a feature importance of 0.135 and a value 0.234 for hispanic participants. With regards to race, there was a larger

impact of the memory column for white and asian participants, -4.019 and -2.961 as opposed to 0.687. This was similar to the results for orientation, which were that it was very important for white and asian participants, being -1.649 and -1.826, as but moderately impactful for Black or African American participants, -0.488.







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Notes about writing:

- the bulk of my writing is not on the specific intersection between fields that I am trying to highlight; the majority of it is explaining all of the context
- since XAI, alzheimer's, applied machine learning in alzheimer's, and fairness are all field with a lot of documentation, I have been heavily relying on getting articles from overview papers

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